

XR-500 Service and Maintenance Manual

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1. Power Requirements

The XR-500 controller must be connected to a **24 V DC power supply**.

The permitted input voltage range is **22 V DC to 28 V DC**. Do not connect the controller to a power source above **28 V DC**.

2. Operating Temperature

The XR-500 is designed to operate at an ambient temperature between **0°C and 50°C**.

Operation outside this range may result in unreliable controller performance.

3. Cooling Requirements

Maintain a minimum clearance of **100 mm** around the ventilation openings.

The cooling fan should remain unobstructed while the controller is operating.

4. Maintenance

Inspect the cooling fan and ventilation openings every **500 operating hours**.

Clean accumulated dust and debris during each inspection.

5. Flow Rate

During normal operation, maintain the process flow rate between **8 L/min and 12 L/min**.

Do not operate the system below **8 L/min** during continuous operation.

6. Safety

Disconnect the 24 V DC power supply before opening the controller enclosure.

Wait at least **5 minutes** after disconnecting power before servicing internal components.